

Streaming RAG for Live AI Assistants

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Abstract. We present a IoT health monitoring platform designed to connect patients and healthcare professionals through real-time data tracking. This system utilizes a hybrid architecture that combines edge computing for local data processing with cloud based storage and analytics. Central to the platform is an AI clinical assistant that leverages on local large language models to interpret biometric data, explain medical symptoms, and provide actionable health advice. Users can interact with the system via dashboards that display vital signs like heart rate and oxygen levels while offering diagnostic insights by means a chat. The paper focuses on the different functionalities of the system, e.g., use of local LLM models, ability to operate in offline modes, real time data streaming end analysis.

Keywords: Local LLMs, Real-time data, AI assistants

1 Introduction

Retrieval-Augmented Generation (RAG) systems have significantly improved the factual grounding capabilities of large language models by combining generative reasoning with external knowledge retrieval. However, most existing RAG architectures are designed for static or slowly changing knowledge bases, where documents are indexed offline and updated periodically. This assumption limits their applicability in real-world environments characterized by continuously evolving information streams.

Many modern applications require AI assistants capable of accessing and reasoning over live data in real time. In these scenarios, information freshness becomes critical, as outdated or hallucinated responses may lead to incorrect decisions, operational inefficiencies, or reduced user trust.

At the same time, integrating streaming data into RAG pipelines introduces several open challenges, including low-latency retrieval, continuous indexing, temporal consistency, adaptive context management, and scalability under high-frequency updates. Existing evaluation methodologies also remain largely focused on static benchmarks and do not adequately capture the requirements of real-time AI systems.

This motivates the need for new retrieval architectures and orchestration strategies that enable language models to operate reliably over dynamic knowledge environments. By exploring real-time RAG systems, this work aims to contribute toward the development of grounded AI assistants that are both contextually accurate and temporally aware.

1.1 Research Questions

- RQ1: What is the optimal architecture for streaming RAG?
- RQ2: Can RAG systems reason over evolving facts?
- RQ3: How should retrieval adapt to continuously changing knowledge?

1.2 Contributions

This paper provides a technical architectural analysis of an hybrid IoT health platform developed at the University of Genoa. The platform is engineered to serve as a critical bridge between patient-side sensors and clinical supervision, facilitating a "Smarter Way to Monitor" through the integration of real-time telemetry and artificial intelligence. The primary objective of the system is twofold: to provide medical experts with a reliable, remote supervision framework and to offer patients an AI-driven clinical assistant capable of interpreting complex vital signs and suggesting immediate, evidence-based actions.

2 Related Work

3 Hybrid Architecture Overview

The platform adopts a distributed hybrid architecture that separates local responsiveness from centralized intelligence across two primary tiers: the **Edge Tier** and the **Cloud Tier**. The current deployment includes a single edge node serving one patient, but the architecture is explicitly designed for horizontal scalability: multiple independent edge nodes can be registered against the same cloud backend, each monitoring a distinct patient and forwarding telemetry through the shared gateway. This allows the system to grow from a single-patient prototype to a multi-patient clinical deployment without architectural changes.

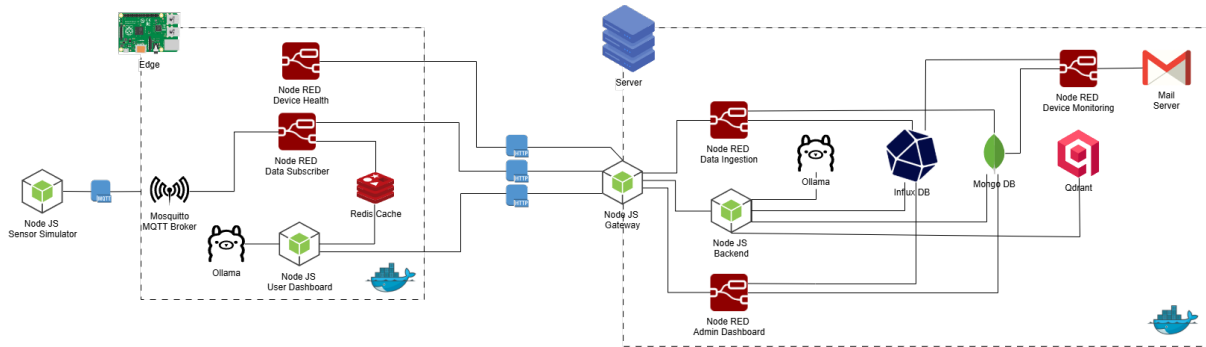


Fig. 1. Hybrid Architecture Overview

4 Edge Tier: Local Processing and Ingestion

The Edge Tier serves as the first processing layer of the platform, handling high-frequency data acquisition, local validation, and direct patient interaction. Sensor readings are filtered and cached locally for immediate use, and simultaneously forwarded to the cloud tier. An integrated AI assistant dashboard allows the patient to query clinical insights in real time, with offline mode support when network connectivity is unavailable. The tier comprises the following components:

- **Node.js Sensor Simulator:** Generates synthetic vital sign data to model clinical scenarios, including healthy baselines and critical events such as cardiac arrest.
- **Mosquitto MQTT Broker:** Acts as the message broker that receiving and distributing data published by the sensor simulator to downstream subscribers.
- **Node-RED Device Health Flow:** Monitors the operational status of connected sensors and devices, detecting and reporting faults or connectivity anomalies.
- **Node-RED Data Subscriber Flow:** Subscribes to the MQTT telemetry stream, validates incoming packets, and forwards them both to the local Redis cache and to the cloud gateway.
- **Redis Cache:** Provides state management for vital signs, ensuring that the most recent physiological readings are immediately available to the local dashboard.
- **Node.js User Dashboard:** Renders real-time patient vitals and exposes the AI assistant chat interface, operating directly against the local Redis cache to support the offline mode.
- **Local Ollama Instance:** Executes a local LLM at the edge to perform intent detection, acting as a safety gatekeeper that determines whether a user query is clinically relevant before escalating it to the cloud, thereby reducing latency and filtering unsafe or out-of-scope requests.

The edge configuration additionally supports a dedicated offline mode, enabling continued local monitoring and dashboarding during periods of network unavailability.

5 Cloud Tier: Centralized Intelligence

The Cloud Tier concentrates the computationally intensive and stateful responsibilities of the platform. It receives validated telemetry from the edge and persists it to specialized databases for long-term trend analysis and clinical review. It exposes a REST API consumed by both the patient-facing dashboard and the AI assistant, hosts the full RAG pipeline for grounded clinical response generation, and coordinates device health monitoring with automated alert notifications to healthcare personnel. The tier comprises the following components:

- **Node.js Gateway:** The central entry point for edge-to-cloud communication via HTTPS, responsible for request validation, and traffic routing to downstream services.
- **Node.js Backend:** The core application server that implements the business logic and exposes the REST API consumed by dashboards and the AI assistant.
- **Node-RED Data Ingestion:** Standardizes incoming telemetry streams from the edge before committing them to the time-series database.
- **Node-RED Device Monitoring:** Orchestrates the device health-state tracking flow and triggers alert notifications when anomalous conditions are detected.
- **Node-RED Admin Dashboard:** Provides a dedicated monitoring interface for system administrators, enabling real-time visibility into data flows and service health.
- **Ollama Instance:** Hosts the full-capacity LLM for complex clinical reasoning and RAG-based response generation.
- **InfluxDB:** Time-series database for the long-term persistence of high-frequency sensor vitals, enabling trend analysis and historical review by clinical experts.
- **MongoDB:** Document-oriented database for application metadata, user profiles, and clinical session records.
- **Qdrant:** Vector search engine that indexes medical document embeddings, enabling semantic retrieval for the RAG pipeline.

6 System Workflows

The platform orchestrates three primary data flows that span both tiers: the IoT telemetry ingestion pipeline, the AI assistant RAG pipeline, and the rule-based diagnosis workflow. Each flow is described below.

6.1 IoT Data Ingestion Workflow

Sensor telemetry is collected at the edge, validated locally, and progressively forwarded to the cloud for persistent storage. The flow ensures that only well-formed data reaches the time-series database, with a dual-validation stage at both tiers.

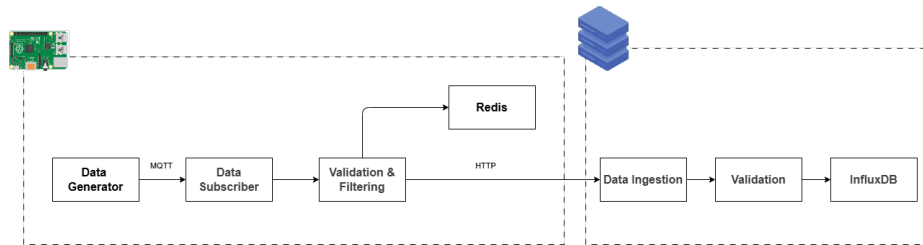


Fig. 2. IoT Data Ingestion Flow

- **Data Generation:** The Sensor Simulator publishes synthetic vital sign telemetry over MQTT.
- **Edge Subscription and Validation:** The Node-RED Data Subscriber flow captures incoming packets, applies validation rules, and caches valid readings in Redis for immediate dashboard use.
- **Cloud Forwarding:** Validated packets are transmitted via HTTPS to the Cloud Gateway.
- **Cloud Ingestion and Validation:** The server-side Node-RED flow applies a secondary validation pass to enforce data integrity before persistence.
- **Persistent Storage:** Clean data is committed to InfluxDB for long-term trend analysis and clinical review.

6.2 AI Assistant and RAG Pipeline

This workflow implements a classical document-based RAG pipeline. The knowledge base consists of biomedical reference texts pre-loaded into the Qdrant vector store. This pipeline does not analyse live sensor data: its sole purpose is to answer patient queries by retrieving relevant passages from the indexed literature and generating evidence-grounded advice. Patient queries are first screened at the edge for clinical relevance, then routed to the cloud where the retrieval and generation steps are executed.

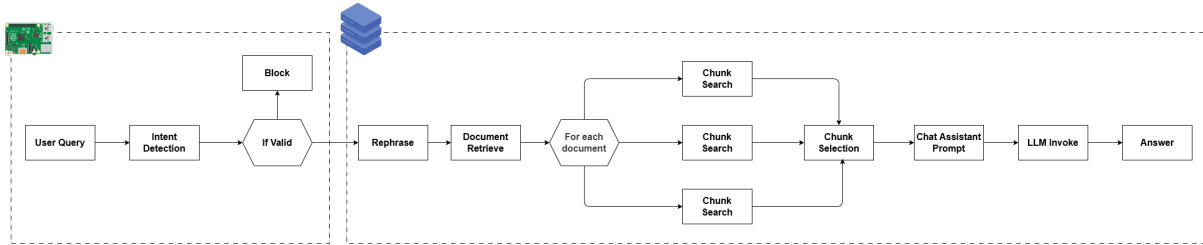


Fig. 3. RAG Pipeline Flow

- **Intent Detection (Edge):** The local edge Ollama instance acts as a safety gatekeeper, confirming the query is clinically relevant before it is escalated to the cloud.
- **Query Rephrasing (Cloud):** The query is rewritten to maximise semantic similarity against the indexed documents.
- **Document Retrieval:** Qdrant returns the most relevant text fragments from the pre-loaded biomedical knowledge base.
- **LLM Invocation:** The cloud Ollama instance generates a response conditioned on both the retrieved documents and the patient’s original query.
- **Answer Delivery:** The evidence-grounded response is streamed back to the patient dashboard.

6.3 Rule-Based Diagnosis Workflow

This workflow extends the standard RAG pipeline with a deterministic Diagnosis Block inserted before the LLM invocation. Rather than relying on a single instantaneous reading, the block aggregates sensor measurements collected over a configurable trailing window — set to five minutes in the current deployment — and computes per-metric statistics (min, max, mean). This window is long enough to smooth transient noise yet short enough to reflect the patient’s current clinical state. The resulting statistics are evaluated against established medical thresholds by a rule engine, which assigns a structured clinical label (e.g., *Tachyarrhythmia*, *Hypoxemia*). This label is then combined with the retrieved document context into a single enriched prompt, ensuring the LLM response is simultaneously grounded in authoritative literature and in the patient’s objective physiological state.

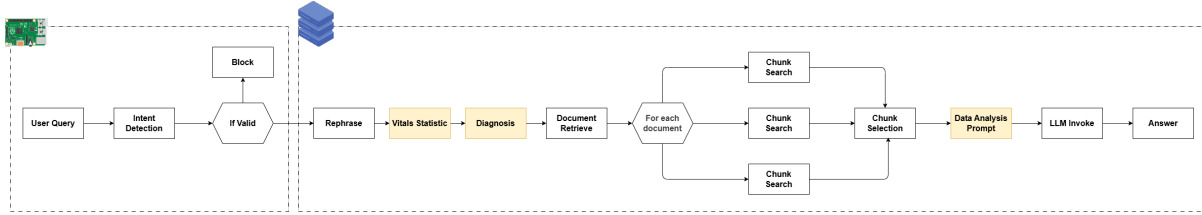


Fig. 4. Data Analyst and Rule-Based Diagnosis Flow

- **Intent Detection (Edge):** As in the standard RAG pipeline, the local Ollama instance screens the query for clinical relevance before escalation.
- **Query Rephrasing (Cloud):** The query is rewritten to maximise semantic similarity against the indexed documents.
- **Vitals Aggregation:** In parallel, sensor readings from the last five minutes are retrieved from InfluxDB and summarized into min, max, and mean values per metric.
- **Diagnosis Labelling:** The rule engine evaluates the aggregated statistics against clinical reference ranges and assigns a structured label with severity metadata.
- **Document Retrieval:** Qdrant returns the most relevant text fragments from the pre-loaded biomedical knowledge base.
- **Data Analysis Prompt:** The diagnosis label and the retrieved passages are combined into a single enriched prompt for the LLM.
- **LLM Invocation and Answer Delivery:** The cloud Ollama instance generates a response conditioned on both sources and streams it back to the patient dashboard.

7 Clinical Dashboard and Simulation Analysis

The platform exposes two distinct user interfaces tailored to their respective roles and deployment tiers: a patient-facing dashboard running at the edge, and an expert dashboard hosted in the cloud.

7.1 Patient Dashboard

The Patient Dashboard runs on the edge node and serves as the primary interface for the monitored patient. It reads vital signs directly from the local Redis cache, enabling real-time visualization of heart rate, blood pressure, oxygen saturation, and respiratory rate even during cloud connectivity loss. The same interface exposes the AI assistant chat, through which the patient can submit clinical queries and receive evidence-grounded responses generated by the RAG pipeline.

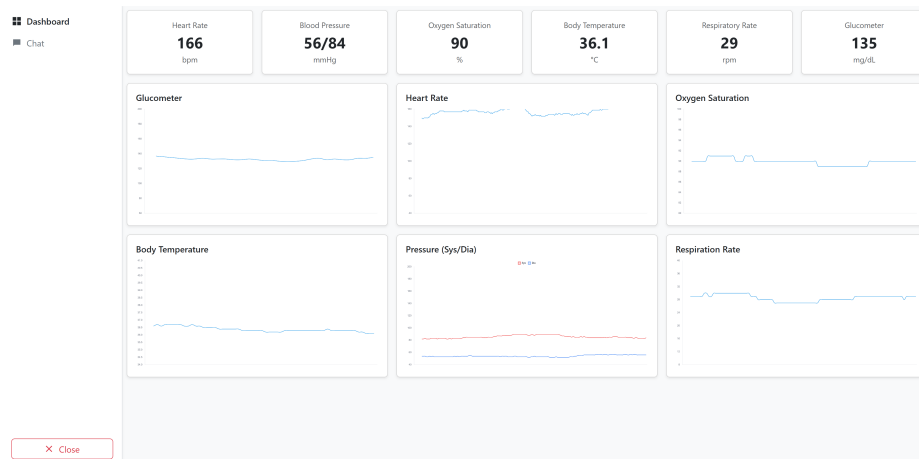


Fig. 5. Patient Dashboard: Vitals Visualization

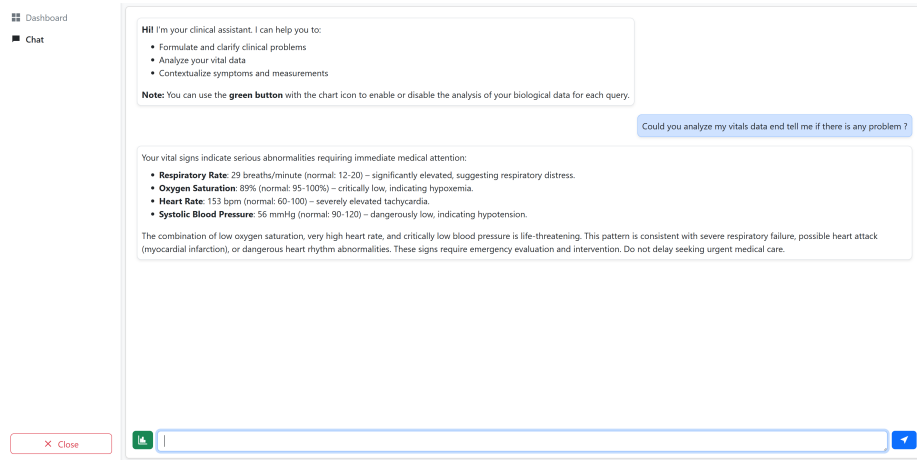


Fig. 6. Patient Dashboard: AI Assistant Chat Interface

7.2 Expert Dashboard

The Expert Dashboard is hosted in the cloud and targets healthcare professionals and system administrators. It provides three main views: a device telemetry panel listing all registered edge nodes and their connectivity status, a session management panel enumerating active and historical patient monitoring sessions, and a session detail view that streams live vitals for a selected patient alongside the full interaction history. This separation allows clinical staff to oversee multiple patients simultaneously and intervene when anomalous conditions are flagged.

Device Id	Name	Status	Last Update
device-1	Device-1	●	2025-12-02T22:29:30.981Z
device-2	Device-2	●	-
device-3	Device-3	●	-
device-4	Device-4	●	-

Fig. 7. Expert Dashboard: Devices Telemetry

Sessions				
Search				
Id	Device Id	Created At	Updated At	Status
692f541f7676d34c8d7b87	device-1	2025-12-02T21:03:27.201Z	2025-12-02T21:03:27.201Z	
692f53e17676d34c8d7b85	device-1	2025-12-02T21:02:25.433Z	2025-12-02T21:03:25.870Z	
692f126c3ab92063c48e2507	device-1	2025-12-02T16:23:08.460Z	2025-12-02T16:27:37.269Z	
692f10233ab92063c48e2506	device-1	2025-12-02T16:13:23.677Z	2025-12-02T16:23:06.825Z	
692f0b423ab92063c48e2505	device-1	2025-12-02T15:52:34.336Z	2025-12-02T16:13:23.674Z	
692f0b273ab92063c48e2504	device-1	2025-12-02T15:52:07.228Z	2025-12-02T15:52:33.436Z	
692ec61f505b94edfe6651c9	device-1	2025-12-02T10:57:35.199Z	2025-12-02T15:52:07.223Z	
692ec5aa505b94edfe6651c7	device-1	2025-12-02T10:55:38.849Z	2025-12-02T10:57:34.354Z	
692ec4cd505b94edfe6651c6	device-1	2025-12-02T10:51:57.716Z	2025-12-02T10:55:38.008Z	
692ec3fb505b94edfe6651c4	device-1	2025-12-02T10:48:27.131Z	2025-12-02T10:51:56.577Z	
692e1535a252cd39bc591bea	device-1	2025-12-01T22:22:45.094Z	2025-12-02T10:48:27.128Z	
692e11ff386e1f7db71be970	device-1	2025-12-01T22:09:03.277Z	2025-12-01T22:22:43.804Z	
692e0f89ca34c424ef51e6d6	device-1	2025-12-01T21:58:33.346Z	2025-12-01T22:09:01.701Z	
692e0e83ce34c424ef51e6d3	device-1	2025-12-01T21:54:11.780Z	2025-12-01T21:58:32.045Z	
692e0d60ee85703df9d949192	device-1	2025-12-01T21:49:20.887Z	2025-12-01T21:54:10.457Z	
692e0bbcf8f237ad81a320d	device-1	2025-12-01T21:42:23.084Z	2025-12-01T21:49:19.343Z	
692e089e0d78fac9219711e	device-1	2025-12-01T21:29:02.342Z	2025-12-01T21:42:21.678Z	

Fig. 8. Expert Dashboard: Active Sessions List

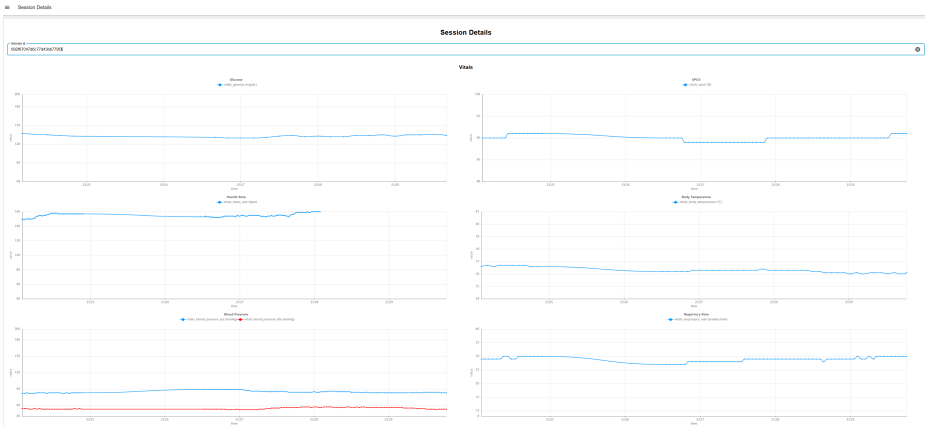


Fig. 9. Expert Dashboard: Detailed Session Vitals

7.3 Simulation Analysis

To validate the system’s diagnostic capabilities, two scenarios were simulated: a healthy baseline and a cardiac emergency. During the “Heart Attack” simulation, the system identified several critical abnormalities:

- **Heart Rate:** 155–166 bpm (Severely elevated / Tachycardia).
- **Blood Pressure:** 57/84 mmHg (Critically low / Hypotension).
- **Oxygen Saturation:** 90–91% (Hypoxemia).
- **Respiratory Rate:** 27–29 rpm (Tachypnea / Respiratory distress).

The AI assistant, informed by the CDS Diagnosis Block, correctly identified these patterns as consistent with Myocardial Infarction and Respiratory Failure, alerting the user to a life-threatening cardiovascular emergency and advising immediate medical intervention.

8 Future Developments (Evolutive)

The strategic technical roadmap focuses on increasing the autonomy, security, and diagnostic capabilities of the platform. As presented in the project slides, the key future evolutions include:

- **LLM Fine Tuning:** Adapting language models specifically for the medical domain to improve response precision.
- **Graph RAG:** Utilizing knowledge graphs to map and retrieve complex medical relationships more effectively than standard vector search.
- **Autonomous Agent:** Deploying AI agents capable of taking autonomous actions or conducting multi-step reasoning.
- **ML Driven Diagnosis:** Transitioning from the current deterministic, rule-based CDS to advanced Machine Learning models for predictive diagnostics.
- **Real Device & Data:** Replacing the sensor simulator with integration for real-world medical devices and actual patient data.
- **Security Improvement:** Hardening the platform to meet strict international health data privacy standards.
- **Kubernetes:** Migrating to Kubernetes for robust container orchestration, high availability, and dynamic scaling.

9 Conclusion

By integrating edge computing for low-latency safety and intent detection with a cloud-based RAG pipeline grounded in authoritative medical literature, this platform provides a robust framework for modern telehealth. The use of deterministic CDS layers alongside probabilistic LLMs represents a significant advancement in clinical safety. Developed at DIBRIS, University of Genoa, this architecture serves as a scalable model for the future of AI-driven remote clinical assistance.

References

1. Author, F.: Article title. *Journal* **2**(5), 99–110 (2016)
2. Author, F., Author, S.: Title of a proceedings paper. In: Editor, F., Editor, S. (eds.) *CONFERENCE 2016, LNCS*, vol. 9999, pp. 1–13. Springer, Heidelberg (2016). <https://doi.org/10.1007/1234567890>
3. Author, F., Author, S., Author, T.: Book title. 2nd edn. Publisher, Location (1999)
4. Author, A.-B.: Contribution title. In: *9th International Proceedings on Proceedings*, pp. 1–2. Publisher, Location (2010)
5. LNCS Homepage, <http://www.springer.com/lncs>, last accessed 2023/10/25